

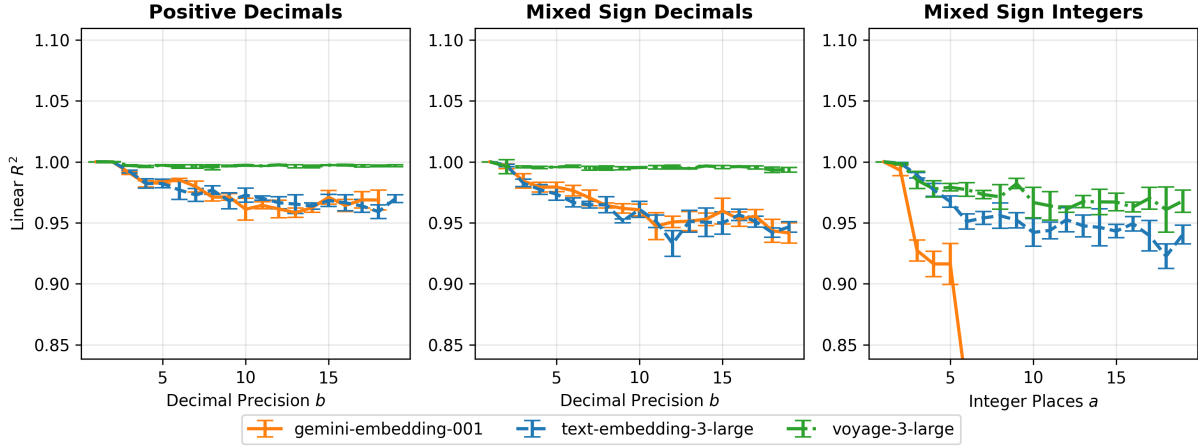


Figure 3: Decimal precision for each dataset plotted against the R^2 score of the linear model reconstructing the original scalars X from their embedded counterparts $\hat{X}$.

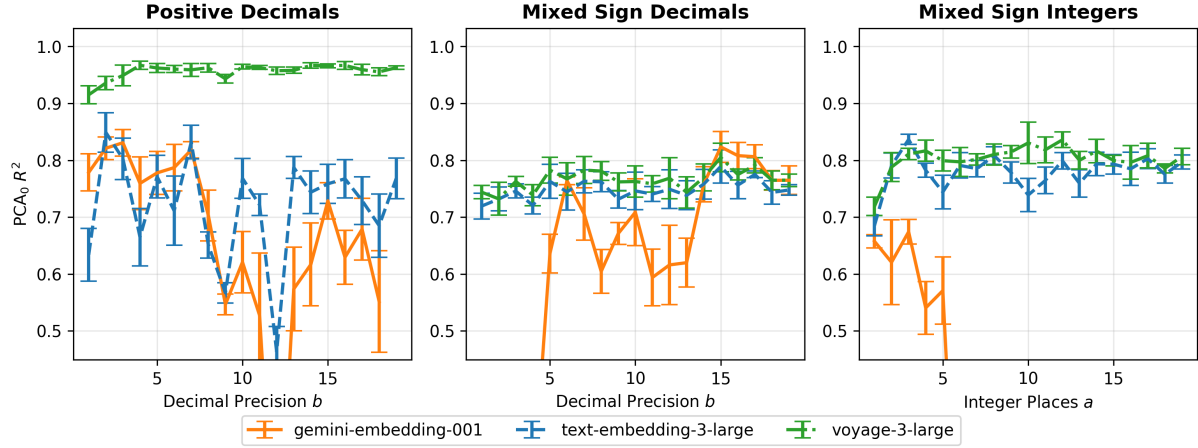


Figure 4: Decimal precision for each dataset plotted against the R^2 of the first component of a PCA projection of the embedded samples $\hat{X}$ against their original counterparts $\hat{X}$.

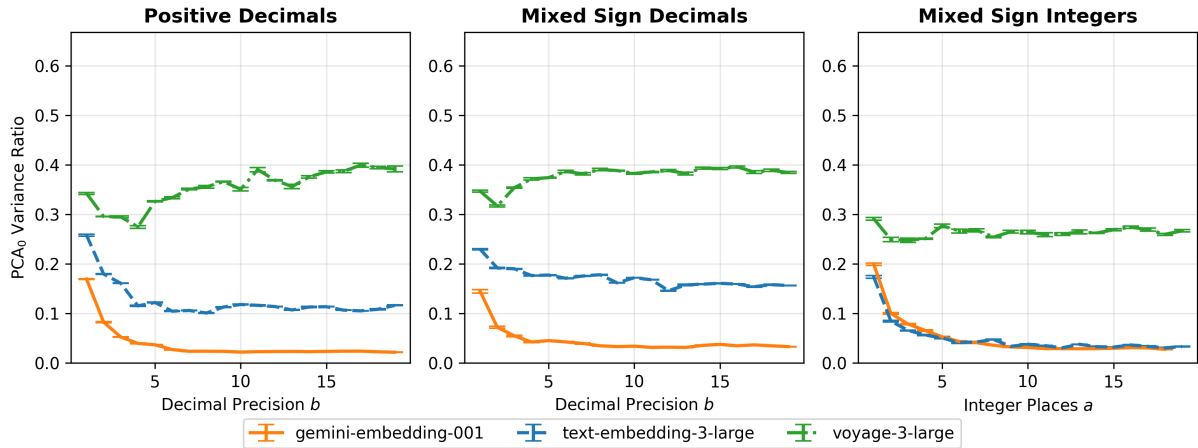


Figure 5: Decimal precision for each dataset plotted against the explained variance ratio of the first component of a PCA projection of the embedded samples $\hat{X}$.

Quantitative results for each dataset and metric can be found in Table 1 for positive decimals, in Table 2 for mixed-sign decimals, and in Table 3 for high-magnitude mixed-sign integers.

Linear Reconstruction

Here we evaluate the relationship in Equation 1, that is, a linear model can perfectly reconstruct samples X from the corresponding samples in the embedding space $\hat{X}$. Figure 3 shows linear R^2 scores plotted against size a and b for each dataset.

On positive decimals performance for all models degrades as the precision (b) increases, though most models maintain $\text{corr}(X, X') \geq 0.95$, indicating that the original samples can be well-constructed from the embedding space even at very high precisions. Introducing mixed-sign decimals degrades performance for all models, most notably for condensed models. OpenAI models in-particular degrade in performance far more with decimal precision when negative decimals are included. Allowing integer places, and larger magnitudes, leads to greater degradation from all models. In-particular, Google’s Gemini-based model drops to at-best medium correlation with precision beyond $a = b = 7$.

Overall we echo the findings of prior work that LLM embeddings of numbers can broadly be used to reconstruct those numbers. However, when numbers are allowed to range in sign and magnitude, such reconstruction suffers. Notably, for true ‘understanding’ of the simple numeric space our models are encoding, there should be little to no variation in these correlations with integer or decimal places.

PCA Correlation

Next, to evaluate the preservation of Equation 3, we measure the correlation of the first PCA component PCA_0 over the embedded datasets $\hat{X}$ against the original scalars they represent. This measures, in effect, whether the embedding correctly encodes the direction of the input samples. Perfect encoders would lead to perfect correlation between PCA_0 and X . We visualise these results in Figure 4.

Performance on positive decimals is highly volatile, with most models dipping with increasing precision into low correlations. Only *voyage-3-large* maintains high correlations against increasing decimal precision, though all models at one decimal place are at least fairly well correlated with the original samples. On both the larger magnitude and mixed sign decimals the Gemini model again performs poorly, though on the mixed sign dataset it recovers some performance with increasing precision. Notably all other models perform comparably on the mixed sign decimal dataset, with little overall degradation as precision increases, and at roughly the same correlation values as on the positive decimals dataset. We discuss these results in more detail in the Discussion, but overall we have shown that the principle component of the embedded samples contains at least most of the information necessary to reconstruct the ordering of X .

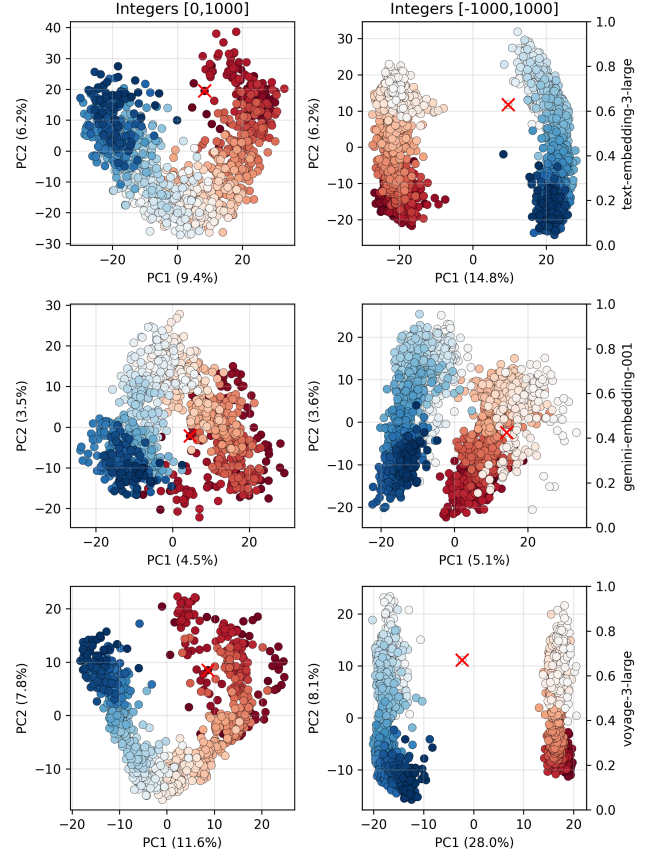


Figure 6: Visualisations of the first two principal components of embeddings of the integers $x \in [0, 1000]$ (left) and $x \in [-1000, 1000]$ (right) for the main OpenAI, Voyage and Gemini models. ‘x’ symbols mark $x = 0$.

PCA Explained Variance

Finally we evaluate the preservation of Equation 2, measuring the relative amount of variance explained by the first principle component PCA_0 . We expect, given the one-dimensional and uniform nature of the input samples X , that PCA_0 explains *all* of the variance in the embedded samples $\hat{X}$. We visualise the explained variance ratio for each model and dataset against increasing precision in Figure 5.

On positive decimals all models show the same exponential-like explained variance decrease with increasing decimal precision, with at most 40% of variance explained at one decimal place. Results are more spread on the mixed sign dataset, with explained variances increasing for most models, with the same pattern on the larger magnitude integer and decimal dataset. Voyage models overall explain more variance in their first principle component, and actually increase in this explained variance as precision increases. Models from OpenAI and Google simply decrease in performance with increasing precision, with the Gemini model explaining the least variance in its first principle component.

Table 1: Metrics for linear models and principal components over a dataset of positive numbers of varying decimal places.

Model	Provider	Linear R^2		PCA R^2		PCA Variance	
		Min	Max	Min	Max	Min	Max
gemini-embedding-001	Google	0.96 ± 0.00	1.00 ± 0.00	0.04 ± 0.04	0.83 ± 0.02	0.03 ± 0.00	0.20 ± 0.00
text-embedding-3-large	OpenAI	0.96 ± 0.01	1.00 ± 0.00	0.46 ± 0.04	0.85 ± 0.03	0.03 ± 0.00	0.17 ± 0.00
text-embedding-3-small	OpenAI	0.96 ± 0.01	1.00 ± 0.00	0.64 ± 0.04	0.87 ± 0.03	0.04 ± 0.00	0.19 ± 0.00
text-embedding-ada-002	OpenAI	0.95 ± 0.01	1.00 ± 0.00	0.25 ± 0.03	0.85 ± 0.02	0.04 ± 0.00	0.17 ± 0.00
voyage-3-large	Voyage	1.00 ± 0.00	1.00 ± 0.00	0.92 ± 0.02	0.97 ± 0.01	0.25 ± 0.00	0.29 ± 0.00
voyage-3.5	Voyage	0.97 ± 0.01	1.00 ± 0.00	0.02 ± 0.02	0.93 ± 0.01	0.14 ± 0.00	0.36 ± 0.00
voyage-3.5-lite	Voyage	0.97 ± 0.01	1.00 ± 0.00	0.64 ± 0.02	0.86 ± 0.01	0.18 ± 0.00	0.33 ± 0.00

Table 2: Metrics for linear models and principal components over a dataset of numbers of varying decimal places and mixed signs.

Model	Provider	Linear R^2		PCA R^2		PCA Variance	
		Min	Max	Min	Max	Min	Max
gemini-embedding-001	Google	0.94 ± 0.01	1.00 ± 0.00	0.01 ± 0.01	0.82 ± 0.03	0.03 ± 0.00	0.14 ± 0.00
text-embedding-3-large	OpenAI	0.93 ± 0.01	1.00 ± 0.00	0.72 ± 0.02	0.79 ± 0.03	0.15 ± 0.00	0.23 ± 0.00
text-embedding-3-small	OpenAI	0.87 ± 0.01	1.00 ± 0.00	0.71 ± 0.02	0.78 ± 0.03	0.18 ± 0.00	0.30 ± 0.00
text-embedding-ada-002	OpenAI	0.85 ± 0.03	1.00 ± 0.00	0.69 ± 0.04	0.77 ± 0.02	0.13 ± 0.00	0.21 ± 0.00
voyage-3-large	Voyage	0.99 ± 0.00	1.00 ± 0.00	0.73 ± 0.03	0.80 ± 0.02	0.32 ± 0.00	0.40 ± 0.00
voyage-3.5	Voyage	0.96 ± 0.01	1.00 ± 0.00	0.73 ± 0.02	0.79 ± 0.02	0.38 ± 0.00	0.45 ± 0.00
voyage-3.5-lite	Voyage	0.91 ± 0.11	1.00 ± 0.00	0.75 ± 0.02	0.80 ± 0.02	0.37 ± 0.00	0.55 ± 0.00

Discussion

Our results reveal a complex picture of how LLM embedding models encode numerical information. First, we demonstrate that embeddings can indeed be used to reconstruct numbers with reasonable fidelity. The linear reconstruction experiments show that most models maintain R^2 scores above 0.95 for simpler numerical datasets, confirming that numerical information is preserved in the embedding space. This finding supports prior work suggesting that language models possess some inherent understanding of numerical relationships.

Second, the first principal component of the embedded representations correlates meaningfully with input precision across most models and datasets. This correlation indicates that the primary axis of variation in the embedding space aligns with the numerical ordering of the input scalars, suggesting that models do capture the fundamental ordinal structure of numbers.

However, our third finding reveals a significant limitation: the explained variance by the first principal component remains consistently low across all models, typically below 40% even for the simplest datasets. This low explained variance is particularly concerning when considered alongside our fourth observation. Given that the first component does correlate well with the original numbers, the low explained variance implies that the embedding space contains substantial additional variation that is not present in the original one-dimensional numerical input.

In Figure 6 we visualise the first two principal components of these principal components for the ‘flagship’ model from each provider over the complete sets of integers $x_+ \in [0, 1000]$ and $x_{\pm} \in [-1000, 1000]$. x_+ shows that while the

first principal component does broadly encode the rank of the original data, the second principal component introduces clear trends unrelated to the original values. Further, all three models on $x_{\pm}$ have the first principal component encoding effectively only the sign of the data, despite the fundamentally continuous nature of the original values. The second component then broadly encodes magnitude, although for the gemini model there is significant ‘bleed’ between the clusters for positive and negative values. In the Appendix we reproduce Figure 6 for increasing number magnitude, and observe that for large magnitudes neither the first nor second principal components represent number magnitude.

This excess variation suggests that embedding models introduce considerable noise into their numerical representations. The high-dimensional embedding spaces capture not only the intended numerical information but also artifacts from the models’ pretraining on diverse text corpora. These artifacts manifest as spurious dimensions of variation that obscure the underlying numerical structure.

Implications

Our findings have several practical implications for applications utilizing LLM embeddings for numerical data. First, numerical understanding appears to degrade significantly with increasing precision, suggesting that simply rounding numbers to fewer decimal places may improve performance in downstream tasks. This finding is particularly relevant for applications requiring numerical reasoning or similarity computation over quantitative data.

Second, the signs of numbers have substantial impact on embedding quality across all tested models. The principal component of variation for all models, with mixed sign val-